



The Law of Self-Maintained Invariance



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The Law of Self-Maintained Invariance sets out a general principle for the persistence of identifiable systems in open environments and builds the Porter Ratio from four postulates. From those postulates, $R = \lambda_{\text{self}} / \lambda_{\text{env}}$ follows as the canonical dimensionless measure of restoration against disruption, $R = 1$ follows as the unique balance point, and above that point inherited organization gains causal continuity and accumulated influence and increasingly constrains what the system becomes next.

Why identifiable systems persist

What has to be true for an identifiable system to persist in an environment that keeps acting on it?

The four postulates

1. **Something has to stay stable.** An identifiable system persists as a distinct entity only if at least one organizational invariant remains stable across interactions with its surroundings. In plain terms: some feature of its organization has to stay the same for the system to still be itself.
2. **The surroundings keep acting.** Open systems undergo continual interactions with their environments that are capable of altering, dispersing, or eroding those organizational invariants.
3. **Persistence requires active counteraction.** When the invariant persists through interactions capable of altering it, organization-preserving processes must counteract that alteration over the relevant interval.
4. **Restoration and disruption compete.** At the effective dynamical level, the net evolution of organizational invariants is determined by the additive competition between internally generated restoration and environmentally generated disruption.

The two rates

- λ_{self} is the rate at which the selected organization is maintained, restored, reinforced, or reliably carried forward. In The Law of Self-Maintained Invariance it is the effective aggregate rate at which internally generated processes reconstruct or preserve organizational invariants inherited from previous system states.
- λ_{env} is the rate at which that same organization is revised, dispersed, overwritten, or disrupted through surrounding interaction. There it is the effective aggregate rate at which environmental interactions alter, disperse, or erase those invariants.

These effective rates provide a complete coarse-grained description of the competing processes governing organizational persistence. In plain terms: for the feature being tracked, these two numbers summarize the whole contest between repair and wear.

The ratio

$$R = \lambda_{\text{self}} / \lambda_{\text{env}}$$

R is the canonical measure of the balance between internally maintained organization and environmental disruption. Both quantities are rates with identical dimensions, so

the ratio stays the same under changes of scale and units. It reads the same whether time is counted in seconds or in years. Repair twice as fast as damage gives $R = 2$. Equal repair and damage give $R = 1$. Damage twice as fast as repair gives $R = 1/2$.

What follows from the postulates

- **$R < 1$:** revision outruns continuation. Organizational invariants undergo net erosion as environmental disruption exceeds internal restoration.
- **$R = 1$:** the two rates balance. This is the unique balance point.
- **$R > 1$:** inherited organization is restored or propagated faster than it is revised. Organizational invariants undergo net accumulation.

Repeated continuation above $R = 1$ allows earlier organization to remain causally active and increasingly constrain what can happen next.

Above $R = 1$, invariants inherited from previous system states are reconstructed more rapidly than they are erased. Accumulated organizational history therefore remains causally active through successive interactions. Earlier organizational states continue constraining later system states, and persistence becomes increasingly determined by accumulated internal history and less by immediate environmental influence.

Substrate independence

The formulation rests on the existence of internally maintained organizational invariants in an open system. It holds for any material substrate or implementation, which makes the law substrate-independent. In plain terms: the comparison does its work wherever both rates can be identified, whatever the system is made of. The result applies to self-maintained organization, persistence, historical constraint, and the accumulation of inherited organizational structure across biological, cognitive, artificial, ecological, and other complex dynamical systems.

Terms to know

- **Organizational invariant.** A feature of a system’s organization that stays stable across its interactions. In plain terms: the part that makes it recognizably the same system.
- **Open system.** A system that keeps interacting with its surroundings.
- **Net erosion and net accumulation.** Whether, over time, the invariant is worn down or built up.
- **Historical constraint.** Earlier organizational states continuing to limit and shape later states.

The bigger picture

This is the first step in measurable form. The Porter Ratio compares the rate at which inherited organization is carried forward with the rate at which surrounding interaction revises it, and $R = 1$ is the balance point.

Full paper on PhilArchive: [The Law of Self-Maintained Invariance](#)

Before this: [Consequential History and the Conditions of Persistence](#).

Next: [The Shape of Persistence](#), where crossing $R^\star$ gives a system an interior form of time.

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